

0	5
---	---

A student investigated the use of cinnamon oil as an antimicrobial substance. She investigated the effect of cinnamon oil on the growth of five different bacterial cultures grown on agar plates.

0	5
---	---

1

The student added 100 mm³ of each bacterial culture from its glass bottle onto a separate agar plate. She spread each bacterial culture evenly over the agar using a spreader.

Describe the aseptic techniques she should use.

[3 marks]

[illegible]

0	5	.	2
---	---	---	---

On each agar plate, the student cut a well (a hole) in the agar.

The well had a diameter of 6 mm. The student added 50 mm³ of cinnamon oil into the well.

Calculate the minimum depth of the well to allow the addition of 50 mm³ of cinnamon oil.

Use the following equation in your calculation:

$$\text{Volume of a cylinder} = \pi r^2 \times l$$

Use 3.14 as the value for π .

Show your working.

[2 marks]

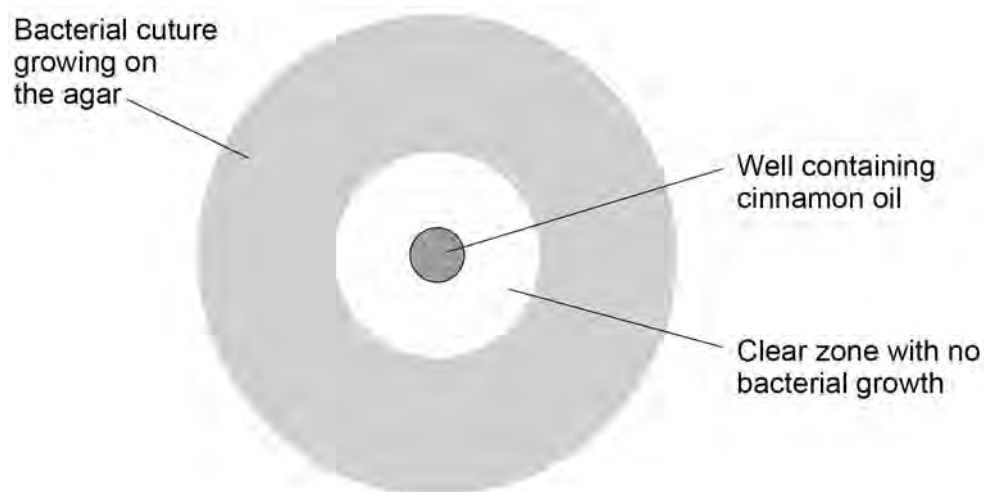
Answer _____ mm

Question 5 continues on the next page

The student kept the plates at 25 °C for 24 hours.

Figure 5 shows what one of her plates looked like after 24 hours.

Figure 5



The student measured the diameter of the clear zone with no bacterial growth around each well. She made these measurements to the nearest whole mm

Table 1 shows her results.

Table 1

Bacterial culture	Diameter of clear zone / mm		
	Cinnamon oil	Positive control	Negative control
<i>Bacillus</i> spp.	15	14	0
<i>Staphylococcus aureus</i>	20	17	0
<i>Listeria monocytogenes</i>	18	12	0
<i>Escherichia coli</i>	16	12	0
<i>Klebsiella</i> spp.	14	12	0
Median for all cultures			0
Mean for all cultures			0
Standard deviation for all cultures	2.4	2.2	0

0 5 . 3

Suggest exactly what the student added to the wells to get the positive control **and** negative control results.

[2 marks]

0 5 . 4

Complete **Table 1** to show the median and mean diameters.

[1 mark]

0 5 . 5

The mean $\pm$ 2 standard deviations includes over 95% of the data.

Use this information to consider whether the standard deviations suggest the differences in means are likely to be due to chance.

Explain your answer, including at least **one** calculation.

[2 marks]

1

Turn over for the next question

Turn over ►

